

Document Use Cases for AI in Projects

Use Case 1: Automated Test Generation

Context: Manual test case creation is time-consuming and inconsistent.

AI Solution: Use an AI tool (e.g., Mabl or Functionize) to generate test cases automatically from user stories, requirements, or code changes.

Benefits:

- Faster test design
- Improved test coverage
- Reduces human error

Example: NLP inputs like “Verify login with invalid password” convert to executable scripts. botgauge.com

Use Case 2: Self-Healing Automated Tests

Context: Frequent UI/DOM changes cause broken tests.

AI Solution: Deploy self-healing frameworks that detect changed elements and update selectors automatically.

Benefits:

- Lower test maintenance cost
- Reduced flaky test failures

Impact: Test stability increases, CI/CD passes more reliably. [Medium](#)

Use Case 3: Predictive Bug Detection

Context: QA effort often not focused on risky components.

AI Solution: Predictive analytics models analyze past defects and code churn to highlight high-risk zones.

Benefits:

- Prioritized testing
 - Early detection of critical defects
- Outcome:** Teams focus effort where it matters most. p99soft.com
-

Use Case 4: Visual Testing

Context: Traditional functional tests miss UI glitches.

AI Solution: Visual AI tools capture screenshots and compare them intelligently.

Benefits: Quick visual regression alerts across devices and browsers. [QA Zen](#)

Use Case 5: AI-Driven Performance Testing

Context: Predicting real user load patterns is hard.

AI Solution: AI simulates realistic usage patterns and suggests bottlenecks.

Benefits: Better performance predictions, optimized capacity planning